



Assignment 2: Analysis of School Performance Using PCA and Clustering

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Course: AI 620 Advance Machine Learning

1. Introduction

The aim of this assignment is to analyse school performance using SAT results and the number of students. The goal is to understand how schools are distributed and to group them into categories using machine learning techniques such as PCA and clustering.

2. Dataset Description and Preprocessing

The dataset contains SAT results for 478 schools, including the number of students and average scores in reading, mathematics, and writing.

Some values were recorded as “s”, which represent missing data. These were replaced with missing values and removed. After cleaning, 421 schools remained.

To simplify the analysis, two new variables were created: total SAT score and average SAT score. These help represent overall performance in a clearer way.

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3. Exploratory Data Analysis

3.1 Distribution of SAT Scores

Figure 1 shows the distribution of total SAT scores.

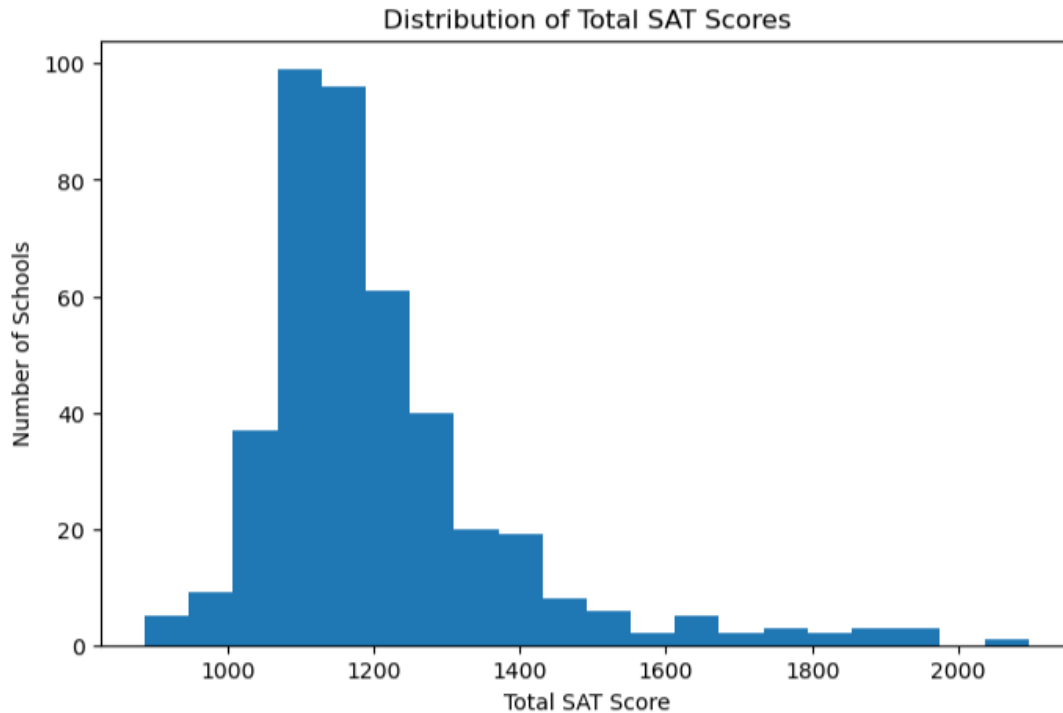


Figure 1: Distribution of total SAT scores across schools

From the histogram, most schools have scores between 1000 and 1300, meaning most schools perform at an average level. Only a small number of schools have very high scores. The average total SAT score is around 1208, with values ranging from 887 to 2096.

3.2 Above and Below Average Schools

To better understand performance, schools were divided into those above and below the average SAT score.

- **145 schools are above average**
- **276 schools are below average**

This shows that most schools are performing below the average level, which suggests that improvement is needed in many schools.

3.3 Distribution of School Size

The number of students also varies between schools. Using the median as a threshold:

- **213 schools are classified as small**
- **208 schools are classified as large**

This shows that the dataset is well balanced in terms of school size.

3.4 Relationship Between Size and Performance

Figure 2 shows the relationship between number of students and performance.

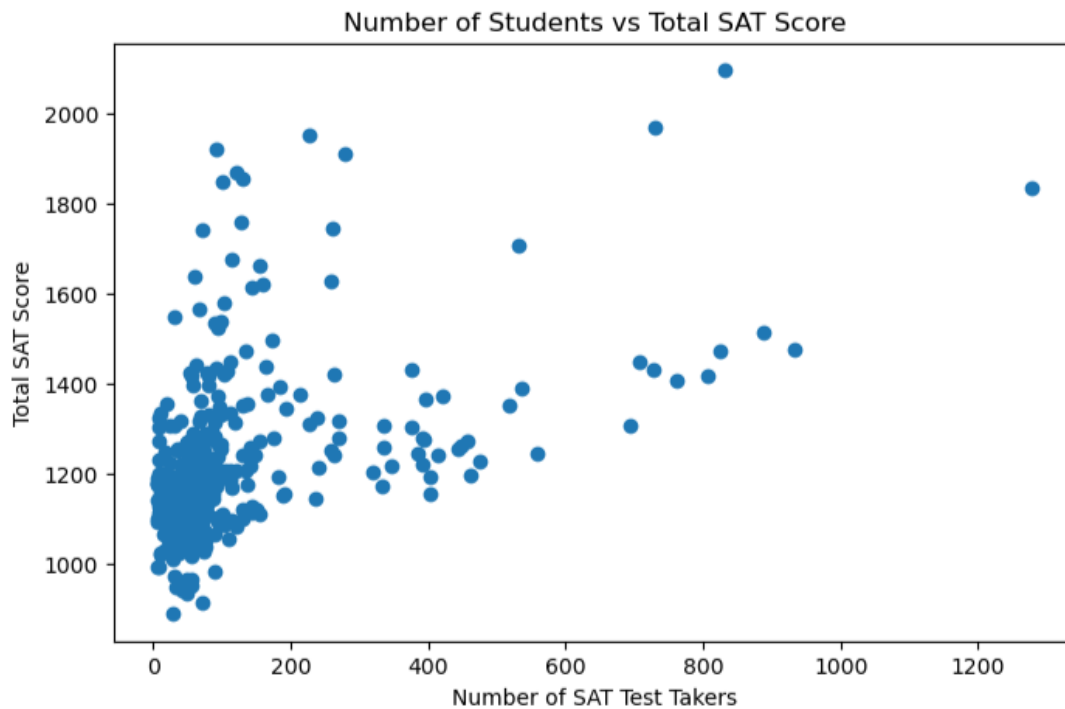


Figure 2: Relationship between number of students and total SAT score

The scatter plot shows no strong relationship. Smaller schools show more variation, while larger schools are closer to average performance.

4. Principal Component Analysis (PCA)

PCA was used to reduce the number of variables and simplify the dataset.

Before applying PCA, the data was standardised using StandardScaler. This is important because the variables have different scales. For example, the number of students can be much larger than SAT scores. Standardisation ensures that all variables contribute equally.

PCA creates new variables called principal components, which are combinations of the original variables. The first component (PC1) captures the main variation and represents overall performance.

Figure 3 shows the PCA result.

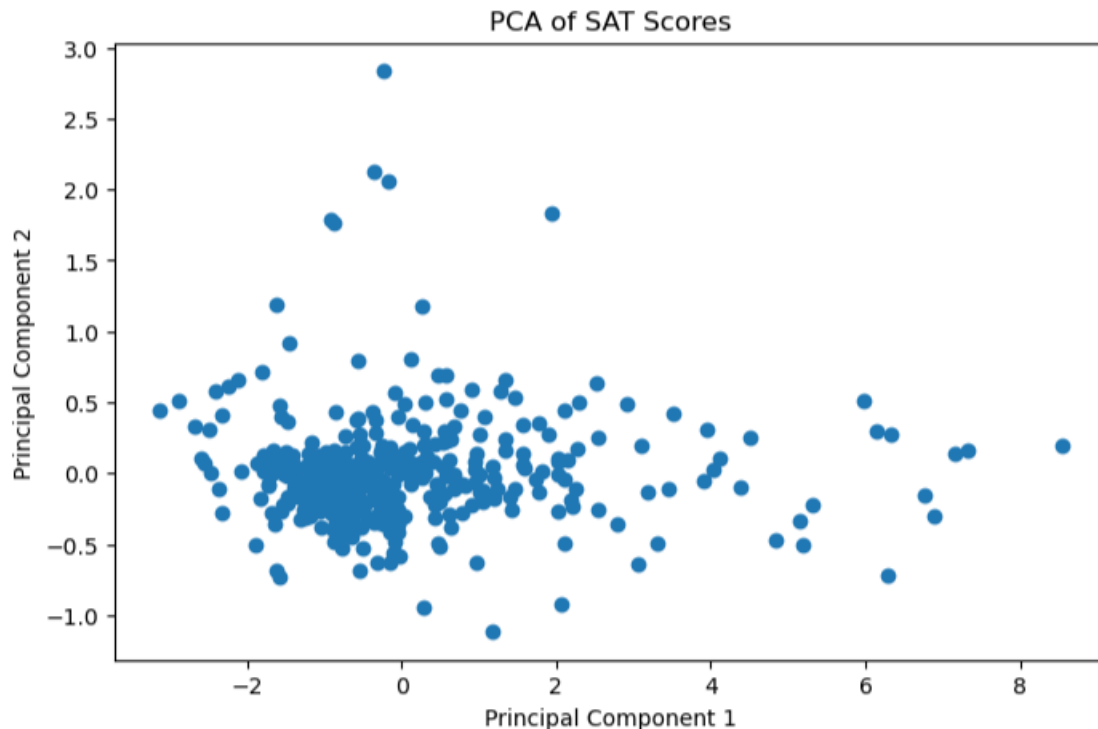


Figure 3: PCA representation of schools based on SAT scores

5. Clustering Analysis

K-Means clustering was used to group schools based on similarity. This method was chosen because it creates clear and easy-to-interpret groups, which is useful for this analysis. It is also a distance-based method, meaning that schools are grouped based on how close they are to each other in the data space. After applying PCA, the data is represented in a simpler form, which makes distance-based clustering more effective. For this reason, K-Means is a suitable choice for grouping schools into categories such as low, medium, and high performance.

5.1 Clustering Based on SAT Scores Only

The first clustering was done using only the SAT scores.

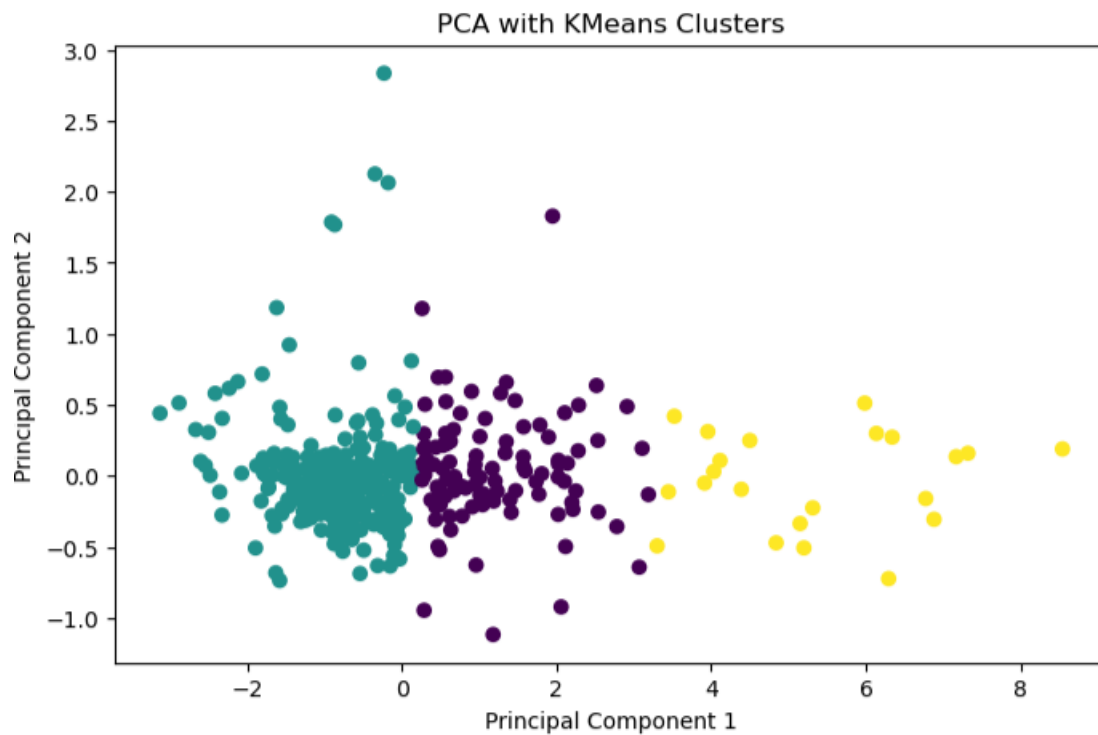


Figure 4: Clustering results based on SAT scores

This grouped schools into low, medium, and high performance, but did not include school size.

5.2 Clustering Including School Size (Final Model)

To improve the analysis, PCA and clustering were repeated including the number of students. This gives more meaningful results:

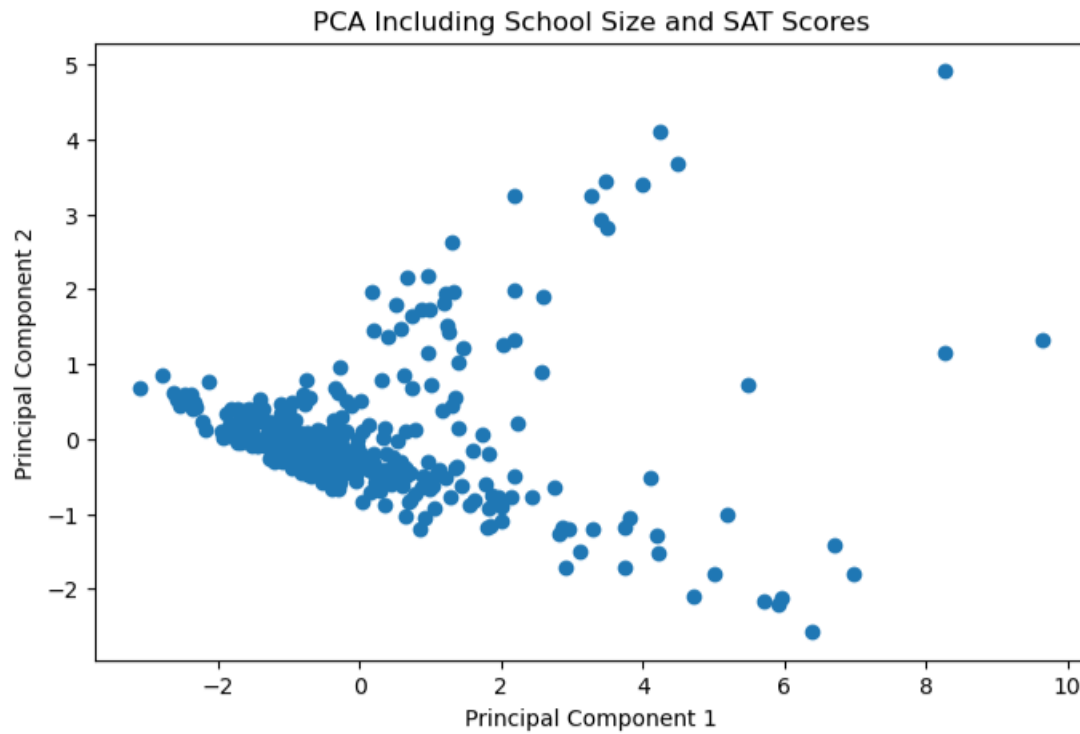


Figure 5: PCA representation including school size

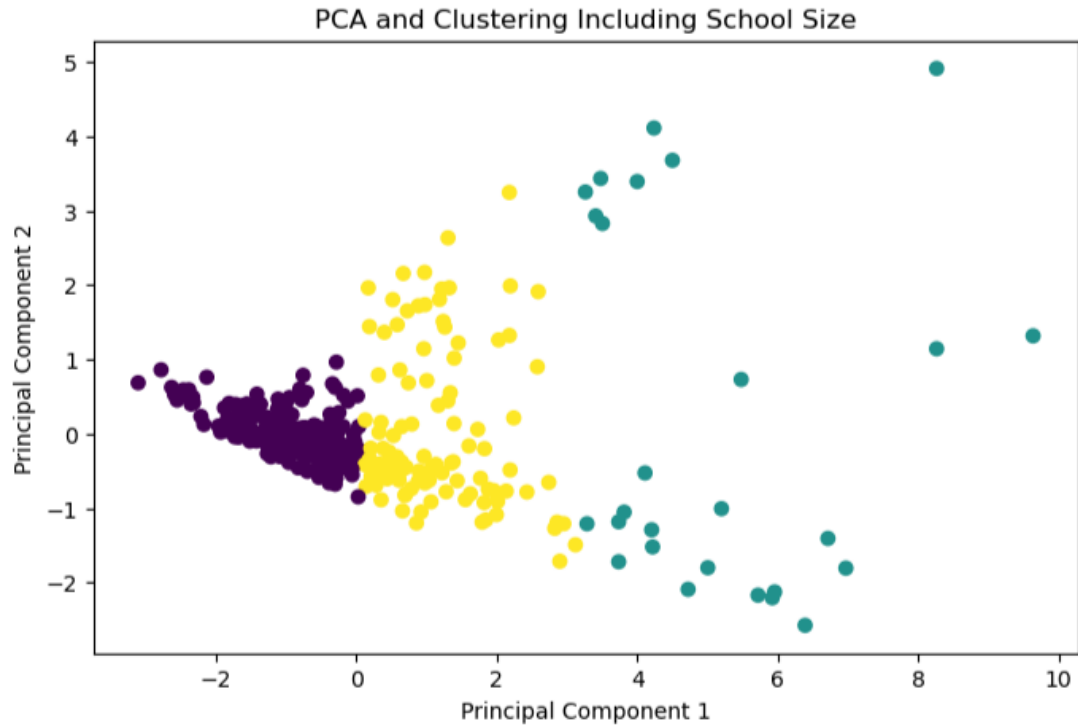


Figure 6: Clustering results including school size and performance

Cluster 0 – Low-performing, small schools

- Average SAT $\approx$ **374**
- Average students $\approx$ **56**
- Number of schools: **284**

Cluster 2 – Medium-performing schools

- Average SAT $\approx$ **438**
- Average students $\approx$ **174**
- Number of schools: **110**

Cluster 1 – High-performing, large schools

- Average SAT $\approx$ **565**
- Average students $\approx$ **423**
- Number of schools: **27**

This clustering combines performance and school size, making it more useful for decision-making.

6. Clustering Evaluation

To check the quality of the clustering, the silhouette score was used.

The result is approximately **0.51**, which indicates a good level of separation between clusters.

Three clusters were chosen because they provide a clear and simple grouping of schools, while still maintaining good clustering quality.

7. Discussion and Insights

From the results, it is clear that most schools are below average, which can be seen both from the histogram and from the above/below average comparison. This suggests that the main issue is not only a few weak schools, but a general performance problem across many schools.

The scatter plot showed that school size alone does not determine performance. However, when size is combined with performance using PCA and clustering, it becomes more meaningful. For example, a large low-performing school represents a bigger problem than a small low-performing school because more students are affected.

The PCA helped simplify the data and allowed the clustering to identify these patterns more clearly. The second clustering model is more useful because it includes both size and performance.

8. Recommendations

Based on the results, different types of schools should be treated differently.

Low-performing schools should receive the most support, since their results show that students may need additional help. This could include better teaching support, resources, and targeted programs.

Medium-performing schools should be improved further, since they are close to becoming high-performing and may need smaller interventions.

High-performing schools should also receive support, but in a different way. They can benefit from advanced programs and opportunities to continue improving.

The most important point is that large low-performing schools should be the highest priority. The reason is that improving these schools will benefit a larger number of students, making the impact of funding more effective.

9. Conclusion

The results showed that most schools are below average and that school size alone does not explain performance. However, combining size and performance provides a better understanding.

The second PCA and clustering model was more useful because it allows decisions to consider both performance and the number of students affected.

Overall, the analysis provides a clear way to categorise schools and supports better decision-making for funding and improvement.

10. Appendix

Use of AI Tools

In this assignment, I used ChatGPT to help me understand concepts like PCA and K-Means, fix code issues, and organise my work.

I am a beginner in coding, so it helped me follow the steps more clearly. I also used it to improve my writing, as I have dyslexia and sometimes find it difficult to structure text.

I reviewed all the results and made sure I understood the final analysis.

Software and Libraries Used

The following software and Python libraries were used in this assignment:

- **Python** (programming language)
- **Pandas** – for data manipulation and analysis
- **NumPy** – for numerical operations

- **Matplotlib** – for data visualisation
- **Scikit-learn** – for machine learning methods, including:
 - StandardScaler
 - PCA
 - KMeans
 - Silhouette Score

Lecture Material

Lecture notes and course materials from the “Advanced Machine Learning” .